Lecture 4

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Credit Risk Overview

- Credit risk is the risk of economic loss due to deterioration in an obligor's creditworthiness
- Losses arise from:
  - Default
  - Restructuring
  - Adverse rating migration
- Credit risk is asymmetric: limited upside, large downside
- Central risk driver in fixed income and credit portfolios

What Is Credit Risk?

- Risk that a borrower fails to meet contractual obligations in full and on time
- Manifests through:
  - Missed or reduced payments
  - Rating downgrades or upgrades
  - Changes in market-required credit spreads
- Exists even without default due to repricing

Three Forms of Credit Risk

- Default Risk
  - Failure to pay interest or principal
  - Results in permanent capital loss
- Ratings Migration Risk
  - Downgrades or upgrades by rating agencies
  - Drives forced selling, spread changes, and price volatility
- Credit Spread Risk
  - Market repricing of credit risk without rating change

Migration Example

- Assume a 5-year, 5.00% coupon, par = 100, annual pay, priced at par (100) today
- Current market yield (YTM) = 5.00%
- Credit deteriorates, so required yield rises to 6.00% (e.g., downgrade / spread widening)
- New price = PV(coupons at 6%) + PV(principal at 6%)
  - PV(coupons) = $5 \times (1-1/1.06^5)/0.06 \approx 5 \times 4.212 \approx 21.06$
  - PV(principal) = $100 / 1.06^5 \approx 100 / 1.338 \approx 74.73$
  - New price ≈ 95.79
  - Mark-to-market loss ≈ 100 − 95.79 = 4.21 points (~4.21%)

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Bond Ratings & Credit Rating Agencies

- Ratings are opinions designed to rank relative credit risk
- These may affect:
  - Initial Pricing (Yield)
  - Secondary Pricing
  - Duration
  - Derivatives (CDS)
- Major agencies: Moody's, S&P Global, Fitch
- Ratings are opinions designed to rank relative credit risk
- Widely embedded in:
  - Mandates and investment guidelines
  - Regulation and capital frameworks
  - Index inclusion criteria

Rating Scales

- Investment grade: AAA/Aaa to BBB-/Baa3
- High yield: BB+/Ba1 and below
- Default categories (e.g., D/C)
- Ratings are ordinal categories, not precise probabilities
- While these are supposed to offer investors an independent and objective view of the risks, they are paid by the issuers so you can judge for yourself how independent and objective these might actually be.

Bond Ratings

	Very High Quality		High Quality		Speculative		Very Poor	
Standard & Poor's	AAA	AA	A	BBB	BB	B	CCC	D
Moody's	Aaa	Aa	A	Baa	Ba	B	Caa	C

At times both Moody's and Standard & Poor's have used adjustments to these ratings: S&P uses plus and minus signs: A+ is the strongest A rating and A− the weakest. Moody's uses a 1, 2, or 3 designation, with 1 indicating the strongest.

Moody's	S&P	
Aaa	AAA	Debt rated Aaa and AAA has the highest rating. Capacity to pay interest and principal is extremely strong.
Aa	AA	Debt rated Aa and AA has a very strong capacity to pay interest and repay principal. Together with the highest rating, this group comprises the high-grade bond class.
A	A	Debt rated A has a strong capacity to pay interest and repay principal, although it is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than debt in higher-rated categories.
Baa	BBB	Debt rated Baa and BBB is regarded as having an adequate capacity to pay interest and repay principal. Whereas it normally exhibits adequate protection parameters, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity to pay interest and repay principal for debt in this category than in higher-rated categories. These bonds are medium-grade obligations.
Ba B Caa Ca	BB B CCC CC	Debt rated in these categories is regarded, on balance, as predominantly speculative with respect to capacity to pay interest and repay principal in accordance with the terms of the obligation. BB and Ba indicate the lowest degree of speculation, and CC and Ca the highest degree of speculation. Although such debt will likely have some quality and protective characteristics, these are outweighed by large uncertainties or major risk exposures to adverse conditions. Some issues may be in default.
C	C	This rating is reserved for income bonds on which no interest is being paid.
D	D	Debt rated D is in default, and payment of interest and/or repayment of principal is in arrears.

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Rating Measurements

What Ratings Are Intended to Measure

- Long-term creditworthiness / capacity to pay
- Through-the-cycle assessment (stability)
- Relative ranking across issuers and sectors

What Ratings Do NOT Measure

- Liquidity risk
- Near-term spread moves
- Event timing (markets often lead)
- Portfolio correlation risk

Issuer Ratings vs Instrument Ratings

- Issuer rating: overall credit quality of obligor
- Instrument rating incorporates:
  - Seniority/subordination
  - Collateral
  - Structural features
- Two bonds of same issuer can have materially different expected recovery

Rating Outlooks and Watches

- Outlook: direction over medium term (positive/stable/negative)
- Watch: near-term event risk (M&A, financing, litigation)
- Watches can move faster than formal rating changes

How Rating Agencies Assess Issuer Creditworthiness

- Ratings express an agency's opinion of an issuer's ability and willingness to meet financial obligations
- Methodologies combine:
  - quantitative financial analysis
  - qualitative assessment of business and governance
  - industry and macroeconomic context
- Ratings are designed to be through-the-cycle, not short-term market signals

Key Inputs and the Rating Committee Process

- Quantitative analysis
  - leverage ratios (e.g., Debt / EBITDA)
  - coverage ratios (e.g., EBITDA / Interest)
  - cash flow metrics (e.g., FCF / Debt)
  - liquidity and refinancing profile
- Qualitative assessment
  - business model stability
  - competitive position and scale
  - management strategy and financial policy
- Rating determination
  - metrics benchmarked against peers and rating guidelines
  - final rating assigned by a rating committee, not a formula
  - outlooks and watches indicate potential rating movement

Why Ratings Differ Across Agencies

- Agencies use different methodologies and weighting schemes
  - emphasis on leverage vs cash flow vs business risk varies
- Treatment of specific factors differs:
  - operating leases
  - pensions and off-balance-sheet liabilities
  - shareholder distributions and financial policy
- Different views on:
  - industry cyclicality
  - competitive advantages
  - downside stress scenarios
- Ratings reflect independent judgment, not consensus
- As a result:
  - split ratings are common
  - differences of one notch are normal

Ratings Strengths and Limitations

Strengths

- Standardized language
- Comparable across issuers and time
- Useful in policy and governance frameworks

Limitations

- Lagging indicator vs market
- Methodology differences across agencies
- Conflicts of interest / issuer-pay model
- Cliff effects (fallen angels, forced selling)

An aside on Duration and Credit Ratings

- Macaulay and modified duration are purely mathematical functions of:
  - Cash-flow timing
  - Coupon
  - Yield to maturity
- A bond's credit rating does not enter the formula.
- Two bonds with identical coupons, maturities, and yields will have the same mechanical duration, regardless of rating.
- However, in practice, ratings materially change the inputs.

An aside on Duration and Credit Ratings (con.)

- Lower-rated bonds trade at higher yields (credit spreads).
- Higher yields:
  - Increase discounting of distant cash flows
  - Shift present value weight toward near-term payments
- For the same maturity and coupon:
  - Lower-rated bonds generally have shorter duration
  - Higher-rated bonds have longer duration

An aside on Duration and Credit Ratings (con.)

- Furthermore, lower-rated bonds embed implicit credit options, even if they are not callable.
  - Examples:
    - Default risk
    - Recovery uncertainty
    - Rating migration
- These features make cash flows state-dependent, which causes effective duration to differ from modified duration.
- Key implications:
  - Price changes reflect both rates and credit spread movements
  - Spread widening during rate shocks dampens price sensitivity
- Effective duration of high-yield bonds is meaningfully shorter than investment-grade bonds with similar maturities

Spread duration vs. interest-rate duration

- For lower-rated bonds, total price sensitivity decomposes into:
  - Interest-rate duration
  - Spread duration
- As rating declines:
  - Interest-rate duration falls
  - Spread duration dominates
- High-yield bonds often behave more like:
  - Short-duration credit instruments
  - Equity-linked risk assets
- This is why high-yield indices have:
  - Low rate sensitivity
  - High sensitivity to growth and risk sentiment

Convexity Effects by Rating

- Convexity effects by rating
- Lower-rated bonds exhibit:
  - Lower convexity
  - Sometimes negative convexity in stress scenarios
- Reason:
  - Higher probability of adverse cash-flow truncation
  - Non-linear default risk
- This further reduces realized duration during market shocks
- Portfolio duration targeting using high-yield bonds often overstates actual rate exposure unless adjusted for effective duration.

Cumulative Default Rates as Inputs

- Historical data often reported as cumulative default rates
- Let τ denote the random time of default
- Cumulative default probability:
  - $Q(t)=P(\tau \leq t)$
- Thus, survival probability:
  - $V(t)=P(\tau > t)=1-Q(t)$
- Historical default studies report $Q(t)$ by rating and horizon
- We now need to convert cumulative defaults into usable default probabilities

Unconditional Default Probability Over One Year

- Consider default during year n:
  - $n-1<\tau\leq n$
- Unconditional default probability:
  - $P(n-1<\tau\leq n)=Q(n)-Q(n-1)$
- Probability of default during the year
- Does not condition on survival into the year

Survival to the Start of the Year

- Probability issuer survives to time $n-1$:
  - $V(n-1)=1-Q(n-1)$
- Only surviving issuers are exposed to default risk in year n
- Conditioning is required to avoid double-counting defaults

Conditional One-Year Default Probability (Discrete Hazard Rate)

- Conditional default probability during year n:

$$\lambda_n = P(n-1 < \tau \leq n \mid \tau > n-1) = \frac{Q(n) - Q(n-1)}{1 - Q(n-1)}$$

- λ_n is the discrete-time hazard rate
- Fundamental building block for:
  - multi-year default modeling
  - expected loss
  - portfolio simulations

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Discrete Default Probabilities Example

- The table below is an example of data produced by rating agencies.
- It shows cumulative default rates during a 10-year period of bonds that had a particular rating at the beginning of the period.
- The probability of a bond defaulting can be calculated from the table using the method just described.
- For example, a bond initially rated B has a probability of 25.895-21.256= 4.639% of defaulting in the fifth year – this is the unconditional probability

Rating	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Aaa	0.000%	0.012%	0.012%	0.037%	0.105%	0.175%	0.245%	0.329%	0.413%	0.497%
Aa	0.022%	0.059%	0.091%	0.159%	0.234%	0.291%	0.348%	0.413%	0.477%	0.542%
A	0.051%	0.165%	0.341%	0.520%	0.717%	0.948%	1.179%	1.468%	1.757%	2.046%
Baa	0.176%	0.494%	0.912%	1.404%	1.926%	2.461%	2.996%	3.514%	4.033%	4.551%
Ba	1.166%	3.186%	5.583%	8.123%	10.397%	12.358%	14.318%	16.200%	18.082%	19.964%
B	4.546%	10.426%	16.188%	21.256%	25.895%	30.184%	34.473%	37.774%	41.076%	44.377%
Caa-C	17.723%	29.384%	38.682%	46.094%	52.286%	56.029%	59.771%	63.639%	67.508%	71.376%

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Discrete Default Probabilities Example

- Now let's calculate the probability the issuer survives to the 4th year:
  - $V(n-1)$= 100- 21.256 = 78.744%
- Thus, λ_5 = (0.04639/0.78744%) = 0.05891
- 5.891% is the *conditional* probability it will default in year 5

Rating	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
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Historical Default Probabilities

- For investment grade bonds, the probability of default tends to increase year over year.
- This is because the possibility of a decline in financial health will increase as more time elapses.
- However, for bonds with a poor credit rating, the probability of default is often a decreasing function of time. **Can anyone tell me why?**

Rating	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7	Year 8	Year 9	Year 10
Aaa	0.000%	0.012%	0.012%	0.037%	0.105%	0.175%	0.245%	0.329%	0.413%	0.497%
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From Discrete Hazard Rates to Default Intensity

- Discrete hazard rates apply over fixed periods (e.g., one year)
- For pricing and term-structure intuition, default risk is often represented in continuous time
- Continuous hazard rates $\lambda(t)$, or default intensity, are :
  - Instantaneous conditional default risk
  - Generalize discrete hazard rates

Survival and Default in Continuous Time (Concept)

- Let $V(t)$ denote survival probability
- In continuous time:

$$V(t) = e^{-\int_0^t \lambda(u)\,du}$$

- Cumulative default probability:

$$Q(t)=1-V(t)$$

- This framework underlies bond pricing and CDS valuation

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Constant Default Intensity (Simplifying Assumption)

- A common assumption:
  - $\lambda(t)=\lambda(\text{constant})$

- Then:
  - $V(t)=e^{-\lambda t}$ $\quad Q(t)=1-e^{-\lambda t}$

- λ represents an average hazard rate over the horizon
- This is a simple baseline assumption that makes default risk easy to model, explain, and calibrate using limited market information.

Recovery Rates

- Recovery R: fraction of exposure recovered after default
- Driven by:
  - seniority and subordination
  - Collateral
  - legal enforceability
  - business cycle
- Recovery affects loss severity, not default likelihood

Loss Given Default & Expected Loss

Loss Given Default:

$$LGD = 1 - R$$

- LGD can vary materially across instruments of the same issuer
- Structure, rights, and regime matter as much as PD

- Expected loss (discrete approximation) over horizon T:

$$EL \approx EAD \times (1 - R) \times Q(T)$$

- Average loss over many realizations
- Priced into credit spreads over time

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Credit Spreads and Default Risk

- Credit spreads compensate investors for:
  - expected default loss
  - systematic credit risk
  - liquidity and technical factors
- Market prices therefore embed implied default risk

Estimating Default Intensity from Bond Prices

- Under simplifying assumptions:
  - constant default intensity
  - constant recovery
  - negligible liquidity premium
- Credit spread approximately satisfies:

$$s \approx \lambda(1-R)$$

or

$$\lambda \approx s/(1-R)$$

Numerical Example: Bond-Implied Default Risk

- Negligible liquidity premia
- Observed spread: s=125 bps
- Assumed recovery: R=40%
- Implied constant hazard:

$$\lambda \approx 0.0125/0.60 = 2.0833\%$$

- Implied 1-year default probability:

$$Q(1) = 1 - e^{-0.020833} \approx 2.0\%$$

Another Angle: 1-Year Risky Zero-Coupon Bond

- Assume you have a zero-coupon bond (not risk-free)
  - Par = 100
  - Risk-free rate r=3% (annual compounding)
  - 1-year risky zero price = 95.92
  - Recovery rate R=40%
  - Negligible liquidity premia
- Cashflows
  - Default during year 1 → receive recovery = 40 at t=1
  - No default → receive par = 100 at t=1

Another Angle: 1-Year Risky Zero-Coupon Bond (con.)

- Pricing equation:

$$95.92 = DF(1)\left[40(1 - V(1)) + 100V(1)\right]$$

$$DF(1) = \frac{1}{1.03}$$

- Solution:

$$V(1) \approx 0.98 \quad \Rightarrow \quad Q(1) = 1 - V(1) \approx 2\%$$

Bond-Implied vs Historical Default Rates

- Historical (physical) default rates:
  - estimated from realized defaults
  - averaged across time and economic cycles
- Bond-implied default rates:
  - inferred from market prices
  - reflect risk premia and liquidity
- The two are not expected to match

Why Bond-Implied Default Rates Are Higher

- Investors require compensation for:
  - systematic default risk
  - correlation during stress
  - Illiquidity
  - Idiosyncratic risk that cannot (practically) be diversified away
- As a result:
  - $\lambda_{bond} > \lambda_{historical}$(typically)

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Physical vs Risk-Neutral Default Probabilities

- Physical (Real-World) Probabilities
  - Measure expected actual default frequencies
  - Estimated from historical data and credit fundamentals
  - Used for carrying out scenario analyses to calculate future potential losses
- Risk-Neutral (Pricing-Implied) Probabilities
  - Implied from market prices of bonds and CDS
  - Adjusted so discounted cashflows equal observed prices
  - Incorporate risk premia for systematic risk, illiquidity, etc.
  - Should be used on valuing instruments, including pricing impact of default risk
- Physical probabilities describe what is expected to happen in the real world.
- Risk-neutral probabilities describe the probabilities required to price credit instruments consistently in financial markets, and therefore embed compensation for risk beyond expected loss.

Why Bond Structure Matters

- Default probability answers “if”
- Bond structure answers “how much is lost”
- Two bonds of the same issuer can have:
  - identical PD
  - very different recovery and risk
- Structural features materially affect:
  - LGD
  - migration risk
  - price volatility

Capital Structure Waterfall

- Claims are paid in strict legal order:
  1. Senior secured debt
  2. Senior unsecured debt (debentures)
  3. Subordinated debt
  4. Preferred equity
  5. Common equity
- Recovery depends on position in the waterfall, not rating alone

Example: Recovery by Seniority

- Assumptions:
  - Firm enters bankruptcy with enterprise value = 100
  - Capital structure:
    - Senior secured debt: 60
    - Senior unsecured debt (debentures): 30
    - Subordinated debt: 20
  - Recovery waterfall:
    - Senior secured: recovers 60 (100%)
      - Remaining value = 40
    - Senior unsecured: recovers 30 (100%)
      - Remaining value = 10
    - Subordinated debt: recovers 10 out of 20 (50%)

Secured vs Unsecured Debt

- Secured debt
  - backed by specific collateral
  - priority claim on assets
  - typically higher recovery
- Unsecured debt
  - no specific collateral
  - relies on issuer’s general credit
  - recovery depends on residual asset value

What Is a Debenture?

- A debenture is an unsecured bond
- Holder has:
  - contractual claim on issuer
  - no lien on specific assets
  - Most corporate bonds are debentures
- Credit risk depends heavily on:
  - overall leverage
  - asset quality
  - covenant protections

Collateral

- Collateral reduces loss severity, not default likelihood
- Key questions:
  - what assets secure the bond?
  - how liquid are they?
  - how are they valued in distress?
  - how enforceable is the lien?
- Correlated asset declines reduce collateral value in stress

Sinking Funds

- Sinking fund = mandatory periodic repayment of principal
- Usually operate in one of two ways
  - Open market purchases
  - Random calls at special price
  - Can escrow funds, but less common
- Reduces:
  - outstanding exposure over time
  - refinancing (maturity wall) risk
- Credit-positive relative to bullet maturity
- Can materially improve recovery and reduce tail loss

Bullet vs Amortizing Structures

- Bullet bond
  - principal repaid at maturity
  - high refinancing risk
- Amortizing bond / sinking fund
  - principal repaid over time
  - lower exposure late in life
- Structure affects both migration risk and LGD

Covenants: Purpose

- Covenants are contractual restrictions protecting creditors
- Designed to:
  - limit risk-shifting by equity holders
  - preserve asset value
  - provide early warning before default
- 3 types:
  - Affirmative
  - Negative
  - Financial

Affirmative (Positive) Covenants

- Affirmative covenants legally require an issuer to take specified actions during the life of the bond
- Purpose:
  - ensure ongoing transparency
  - preserve the issuer's legal and operating capacity
  - allow creditors to monitor credit quality
- These covenants do not limit business strategy directly, but support enforcement of other protections
- Common examples:
  - Provide periodic financial statements and compliance certificates
  - Maintain insurance on material assets
  - Pay taxes and other material obligations when due
  - Maintain corporate existence and required licenses

Negative Covenants

- Negative covenants legally restrict an issuer from taking specified actions without creditor consent
- Purpose:
  - prevent risk-shifting from creditors to equity holders
  - preserve asset value and creditor priority
  - limit increases in leverage and structural weakening
- These are the primary protective covenants in bond documentation
- Common examples:
  - Limitation on additional indebtedness
  - Limitation on liens (protects seniority)
  - Restrictions on asset sales or sale–leasebacks
  - Restrictions on dividends, share buybacks, or distributions

Financial Covenants

- Financial covenants impose quantitative constraints on an issuer's financial condition
- Purpose:
  - provide early warning before payment default
  - force renegotiation while value remains
- Typically expressed as ratios calculated from financial statements
- Two types:
  - Maintenance covenants: must be satisfied at all times or tested periodically
  - Incurrence covenants: tested only when the issuer takes a specific action (e.g., issuing new debt)
- Examples:
  - Maximum leverage ratio
  - Minimum interest coverage ratio
  - Minimum cash on hand

Portfolio Credit Risk Management — Overview

- Portfolio credit risk concerns losses from joint credit events, not just isolated defaults
- Losses arise from:
  - Defaults
  - ratings migration
  - recovery shortfalls
- Portfolio outcomes are driven by co-movement in stress, not average behavior

Why Portfolio Credit Risk Is Different

- Single-name analysis focuses on:
- default probability
- recovery
- Portfolio risk depends on:
  - which names default together
  - how exposures interact in downturns
- Diversification breaks down when defaults cluster